

JEEVA ANANTH V

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Professional Summary

Software Engineer (Machine Learning) specializing in **Generative AI, NLP, and Agentic AI systems**. Experienced in architecting, training, and deploying **LLM-powered, RAG-based enterprise applications** including **KnitiBot**, leveraging **pgvector** and **Kafka**-driven data pipelines for real-time knowledge orchestration. Skilled in production-grade ML engineering involving **model optimization** (TensorRT, ONNX), secure **FastAPI microservices**, and **on-device inference** on Raspberry Pi 5. Demonstrated success delivering measurable performance gains including **20% latency reduction** and **15% accuracy improvement**. Currently advancing **multi-agent MCP-enabled** architectures for emotionally intelligent AI systems.

Core Competencies

Machine Learning & AI: Deep Learning, Transfer Learning, Model Optimization, TensorRT, ONNX

Generative AI & NLP: LLMs (GPT, LLaMA, Falcon), LangChain, RAG, Hugging Face Transformers, Embeddings, Vector DBs (FAISS, Pinecone, Milvus, pgvector), BERT, NER

Computer Vision: YOLOv8, OpenCV, Image Segmentation, Object Detection, Tracking, Anomaly Detection

MLOps & Data Engineering: Kafka, Airflow, Redis, PostgreSQL, Docker, Kubernetes, MLflow, W&B, CI/CD, AWS

Backend Engineering: FastAPI, Flask, REST APIs, OAuth2, Celery, Microservices, API Security

Programming & Data Tools: Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Seaborn, Matplotlib

Professional Experience

Software Engineer

Jul 2024 – Present

Count AI

Coimbatore, India

- Developed **KnitiBot**, an enterprise **multimodal RAG chatbot** backed by **PostgreSQL + pgvector**, integrating internal documents and knowledge resources into a unified interface and significantly reducing manual lookup and overhead across teams.
- Implemented scalable real-time data pipelines using **Apache Kafka** to stream, preprocess, and index data for downstream ML services and RAG-based applications.
- Architected secure **FastAPI + PostgreSQL** microservices with JWT/OAuth2 for model serving, document ingestion, and user management, improving reliability and scalability of production AI systems.
- Contributed to **SuperNova**, a compact AI-powered knitting monitoring machine built on **Raspberry Pi 5** with an AI accelerator HAT, running optimized on-device ML inference for pattern detection and machine control.
- Containerized and deployed ML inference services using **Docker** with scalable production deployment workflows, enabling resilient rollouts, observability, and high-availability model serving.

Research Associate

Feb 2024 – Jun 2024

Terv Pro / Top Freshers

Chennai, India

- Delivered classes and workshops on **Python, Machine Learning, Deep Learning, Data Science, AI, and Generative AI** to over **500+ students** across multiple colleges, contributing to improved technical readiness and campus placement outcomes.
- Implemented proof-of-concept **LLM + RAG** applications aligned with recent ML/AI research papers and designed hands-on NLP and Computer Vision projects to strengthen applied understanding and implementation skills.

Machine Learning Intern

Jun 2023 – Aug 2023

Pantech E-Learning & NSIC

Chennai / Remote

- Implemented ML workflows in TensorFlow and Scikit-learn across diverse datasets, including preprocessing automation and evaluation.
- Studied ML deployment fundamentals and contributed to reproducible experimentation through structured documentation.

Key Projects

Elysium AI – Emotionally Intelligent Multi-Agent Architecture (Open-Source in Progress)	2025
– Engineering a self-evolving, emotionally aware multi-agent system with adaptive behavior based on real-time affective state embeddings. – Integrating a production-focused RAG pipeline + MCP communication for cognitive and emotional context sharing during coordinated decision-making. – Developing new evaluation metrics (affective alignment, resonance mapping) to measure emotional-context effectiveness beyond standard NLP metrics.	
Inquiro – Multimodal RAG Chatbot System	2024
– Developed chatbot integrating PDF, web, and video sources using LangChain + Google Generative AI. – Implemented Retriever-Augmented Generation (RAG) with Chroma Vector DB, improving contextual accuracy by 90%.	
Personalized Medical Advice Chatbot	2024
– Designed chatbot prototype using LLMs + RAG, Pinecone, and Milvus, achieving 1.2s average response latency in evaluations. – Automated data ingestion using Scrapy and deployed test container via Docker + FastAPI on AWS.	
Cogito – Mental Health AI Chatbot (Academic Project)	2024
– Built proof-of-concept NLP chatbot using Hugging Face Transformers with sentiment analysis & entity recognition. – Conducted 500+ simulated interactions to validate contextual empathy and response coherence.	
Hybrid QA System for CUDA Docs (Academic Project)	2023
– Developed hybrid retrieval pipeline combining BERT + BM25 using Scrapy + FAISS + Milvus.	
Perinatal Health Risk Prediction (Academic Project)	2023
– Implemented ML models using Scikit-learn + IBM Watson, achieving 92% F1-score in prenatal health prediction.	

Education

Government College of Engineering, Erode	Aug 2021 – May 2024
<i>B.E. in Electronics and Communication Engineering</i>	<i>Tamil Nadu, India</i>
Rane Polytechnic Technical Campus, Trichy	Jun 2018 – May 2021
<i>Diploma in Mechatronics Engineering</i>	<i>Tamil Nadu, India</i>